

CLUSTERING

CLUSTERING ALGORITHMS – PART VII

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CLUSTERING ALGORITHMS

The following concepts will be introduced:

- ✓ **GRAPH-BASED CLUSTERING**
- ✓ **SPARSIFICATION**
- ✓ **PROXIMITY GRAPH**

HIERARCHICAL CLUSTERING ALGORITHMS take a **graph-based view of data**, in which **data objects are represented as nodes** and the **proximity between two data objects** is represented by the **weight of the edge between** the corresponding **nodes**.

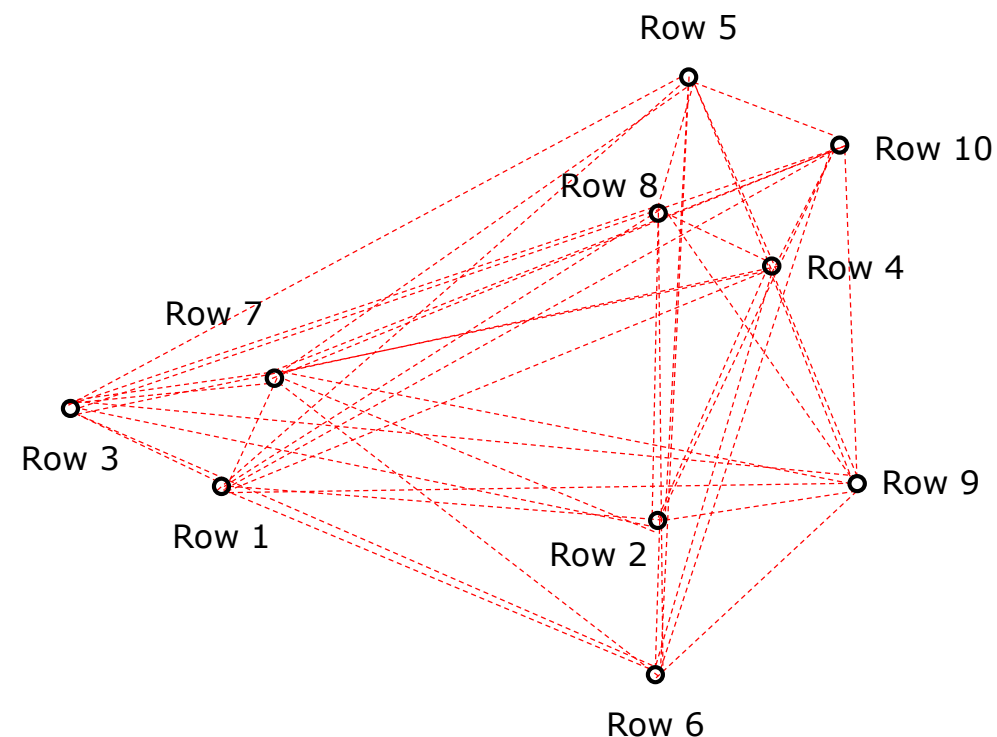
Additional **GRAPH-BASED CLUSTERING** algorithms have been developed which exploit a number of properties and characteristics of graphs. The key approaches are the following:

- **SPARSIFY THE PROXIMITY GRAPH** to keep only the **connections of an object** (record) **with its nearest neighbors**. It is useful for handling noise and outliers, and exploits highly efficient graph partitioning algorithms which have been developed for sparse graphs
- **DEFINE A SIMILARITY MEASURE** between two objects based on the number of **nearest neighbors** that they share. It helps to overcome problems with high dimensionality and clusters of varying density
- **DEFINE CORE OBJECTS** and **BUILD CLUSTERS AROUND THEM**. This requires to introduce a notion of density-based on a proximity graph or a sparsified proximity graph
- **USE THE INFORMATION IN THE PROXIMITY GRAPH** to provide a more sophisticated evaluation of whether two clusters should be merged. Two clusters are merged only if the resulting cluster will have characteristics similar to the original two clusters

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	Row 1	Row 2	Row 3	Row 4	Row 5	Row 6	Row 7	Row 8	Row 9	Row 10
Row 1		0.3	0.8	0.2	0.1	0.2	0.9	0.2	0.1	0.1
Row 2			0.2	0.4	0.1	0.9	0.4	0.4	0.6	0.2
Row 3				0.1	0.2	0.1	0.8	0.1	0.1	0.1
Row 4					0.7	0.5	0.2	0.9	0.5	0.9
Row 5						0.1	0.1	0.8	0.3	0.9
Row 6							0.2	0.2	0.3	0.1
Row 7								0.1	0.1	0.1
Row 8									0.3	0.8
Row 9										0.4
Row 10										

PROXIMITY MATRIX

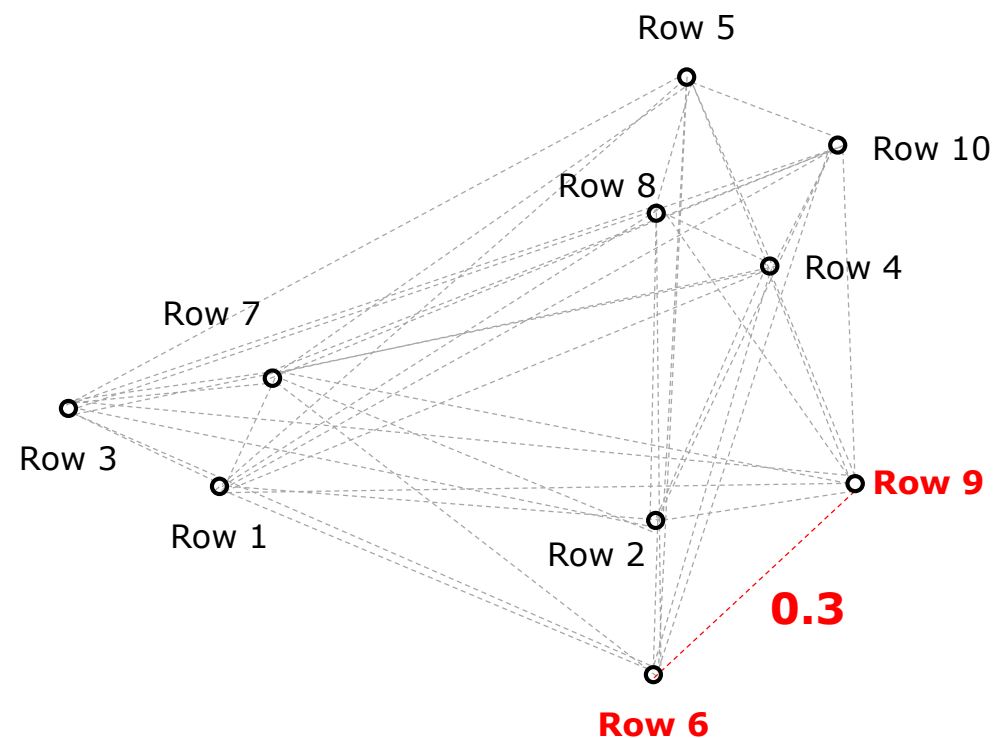


PROXIMITY GRAPH

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Row 2			0.2	0.4	0.1	0.9	0.4	0.4	0.6	0.2
Row 3				0.1	0.2	0.1	0.8	0.1	0.1	0.1
Row 4					0.7	0.5	0.2	0.9	0.5	0.9
Row 5						0.1	0.1	0.8	0.3	0.9
Row 6							0.2	0.2	0.3	0.1
Row 7								0.1	0.1	0.1
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PROXIMITY MATRIX



PROXIMITY GRAPH

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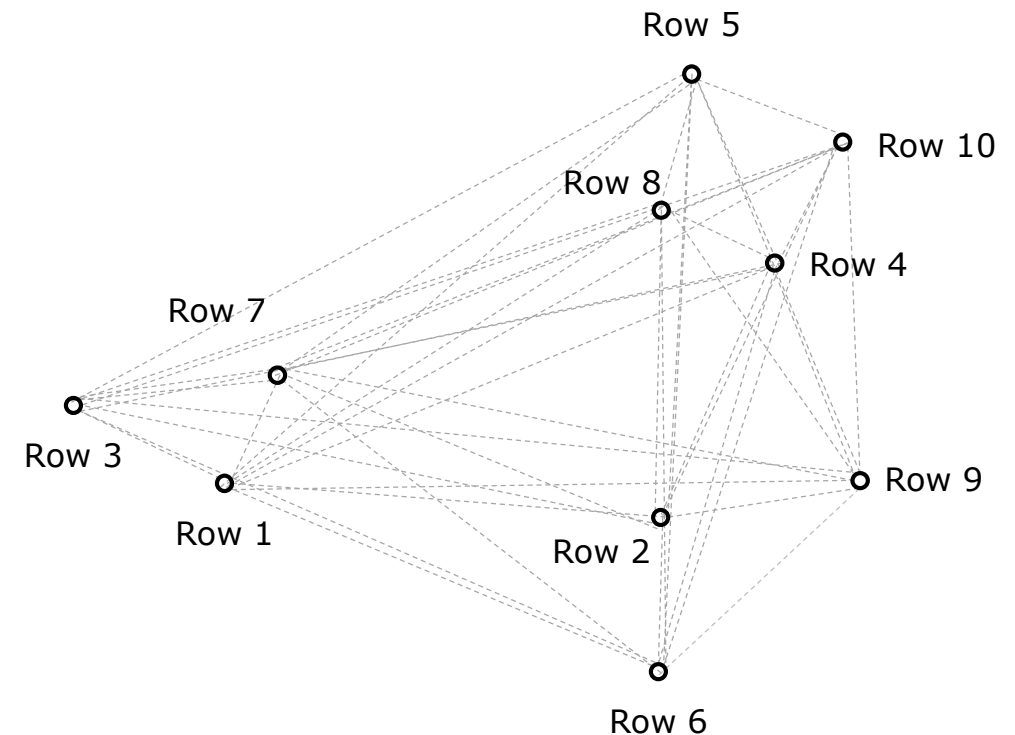


Every object has some level of similarity to every other object (record), for most of data sets, objects are highly similar to a small number of objects and weakly similar to most other objects.

SPARSIFY THE PROXIMITY GRAPH (MATRIX)

Set many of the low similarity values to 0 before beginning the actual clustering process.

Break all links below a threshold or keep only links to the k-nearest neighbors of each point.

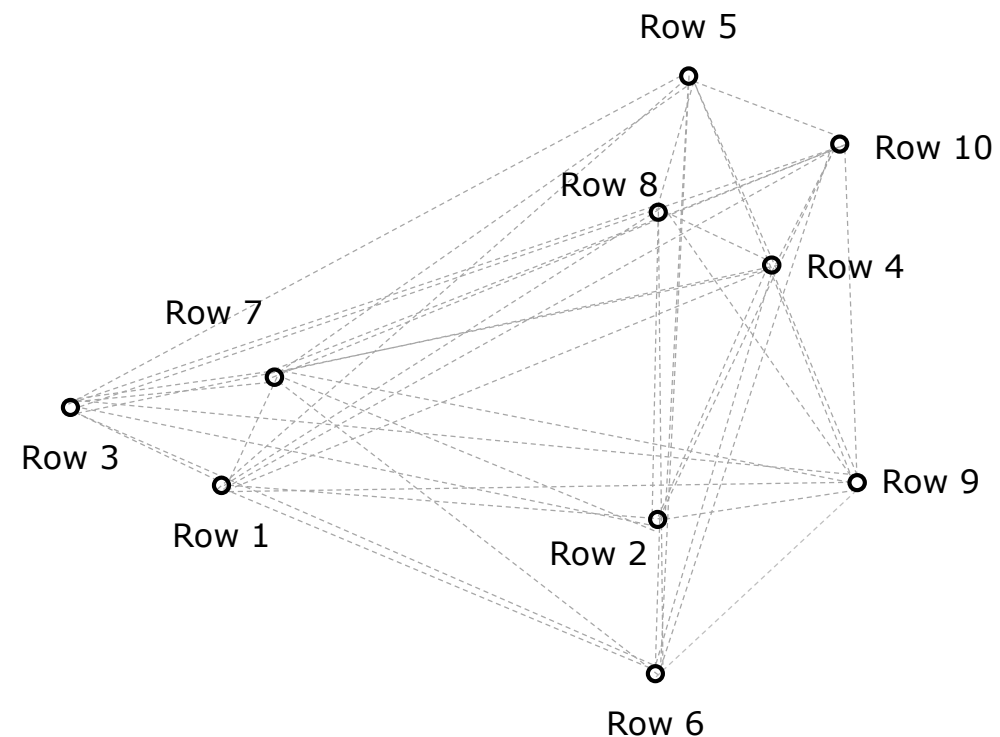


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Row 1			0.8				0.9			
Row 2				0.4		0.9	0.4	0.4	0.6	
Row 3							0.8			
Row 4					0.7	0.5		0.9	0.5	0.9
Row 5										0.9
Row 6										
Row 7										
Row 8										0.8
Row 9										0.4
Row 10										

PROXIMITY THRESHOLD = 0.3

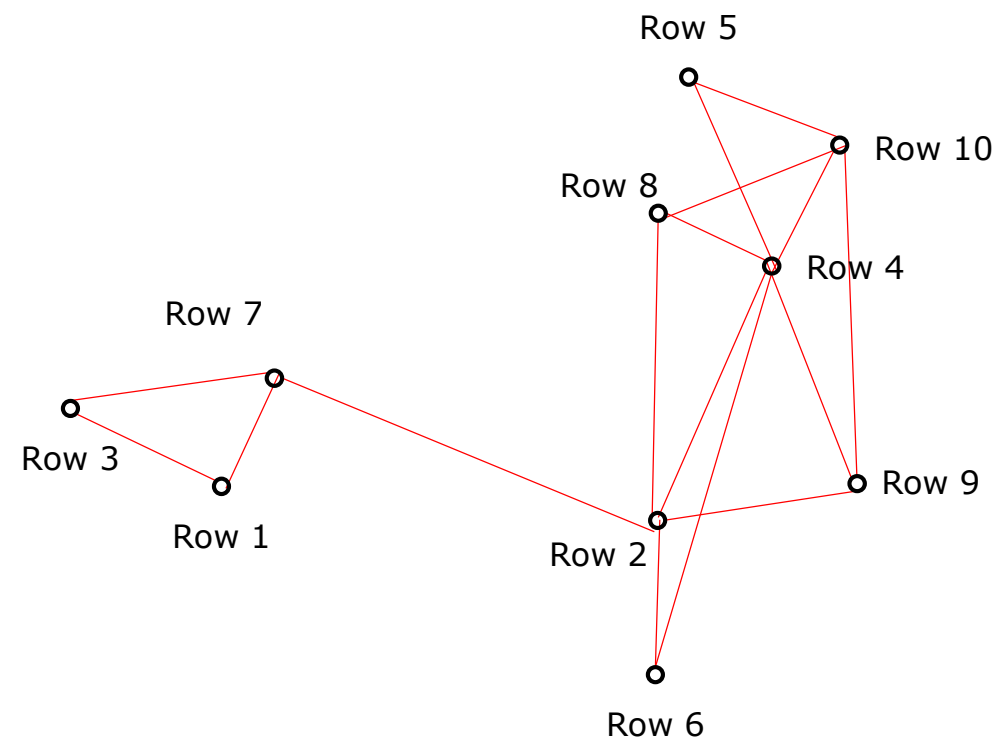


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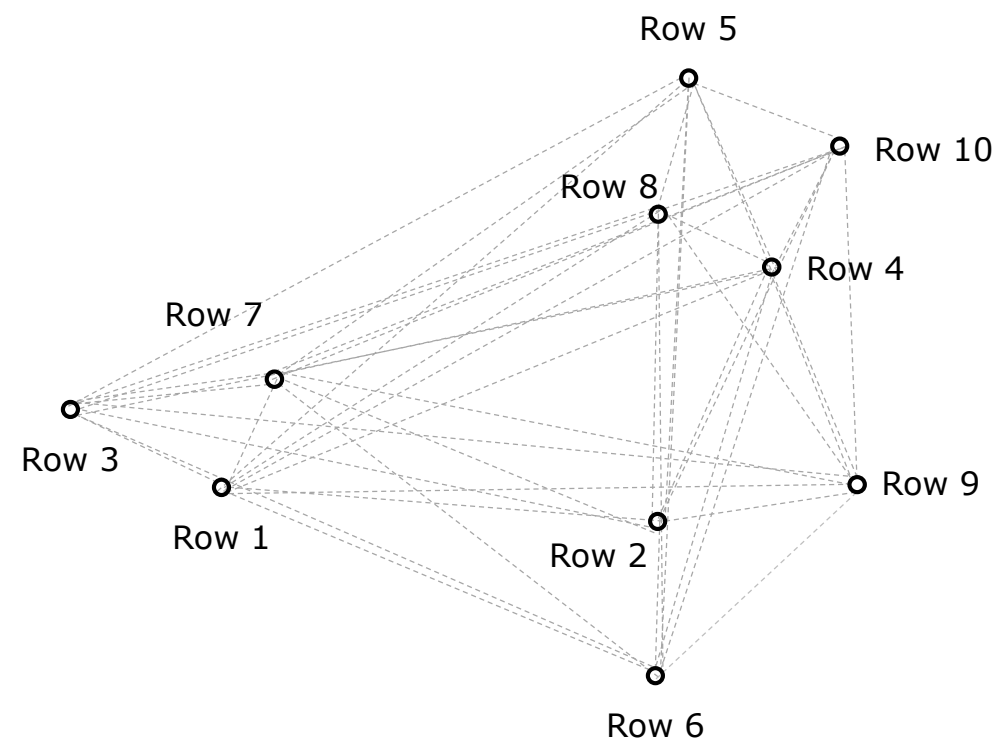
SPARSIFIED PROXIMITY GRAPH

HIERARCHICAL CLUSTERING ALGORITHMS take a graph-based view of data, in which data objects are represented as nodes and the proximity between two data objects is represented by the weight of the edge between the corresponding nodes.

K-NEAREST NEIGHBOR

	Row 1	Row 2	Row 3	Row 4	Row 5	Row 6	Row 7	Row 8	Row 9	Row 10
Row 1		0.3	0.8	0.2	0.1	0.2	0.9	0.2	0.1	0.1
Row 2	0.3		0.2	0.4	0.1	0.9	0.4	0.4	0.6	0.2
Row 3	0.8	0.2		0.1	0.1	0.1	0.8	0.1	0.1	0.1
Row 4	0.2	0.4	0.1		0.7	0.5	0.2	0.9	0.5	0.9
Row 5	0.1	0.1	0.1	0.7		0.1	0.1	0.0	0.3	0.9
Row 6	0.2	0.9	0.1	0.5	0.1		0.2	0.2	0.3	0.1
Row 7	0.9	0.4	0.8	0.2	0.1	0.2		0.1	0.1	0.1
Row 8	0.2	0.4	0.1	0.9	0.0	0.2	0.1		0.3	0.8
Row 9	0.1	0.6	0.1	0.5	0.3	0.3	0.1	0.3		0.4
Row 10	0.1	0.2	0.1	0.9	0.9	0.1	0.1	0.8	0.4	

K = 3



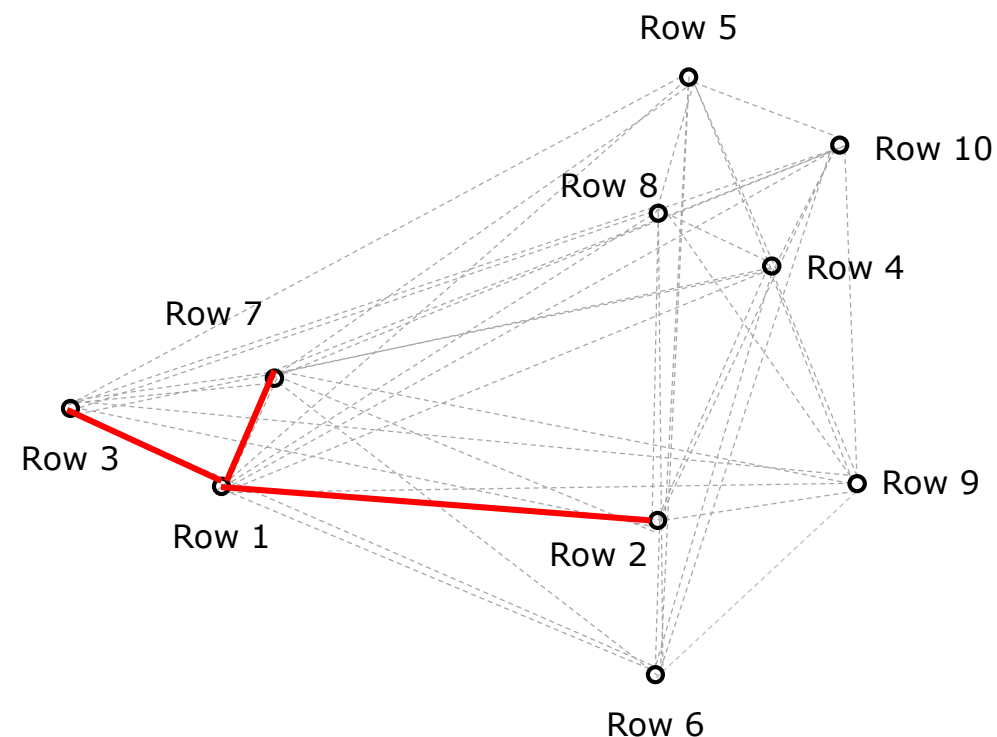
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Row 3	0.8	0.2		0.1	0.1	0.1	0.8	0.1	0.1	0.1
Row 4	0.2	0.4	0.1		0.7	0.5	0.2	0.9	0.5	0.9
Row 5	0.1	0.1	0.1	0.7		0.1	0.1	0.0	0.3	0.9
Row 6	0.2	0.9	0.1	0.5	0.1		0.2	0.2	0.3	0.1
Row 7	0.9	0.4	0.8	0.2	0.1	0.2		0.1	0.1	0.1
Row 8	0.2	0.4	0.1	0.9	0.0	0.2	0.1		0.3	0.8
Row 9	0.1	0.6	0.1	0.5	0.3	0.3	0.1	0.3		0.4
Row 10	0.1	0.2	0.1	0.9	0.9	0.1	0.1	0.8	0.4	

K = 3



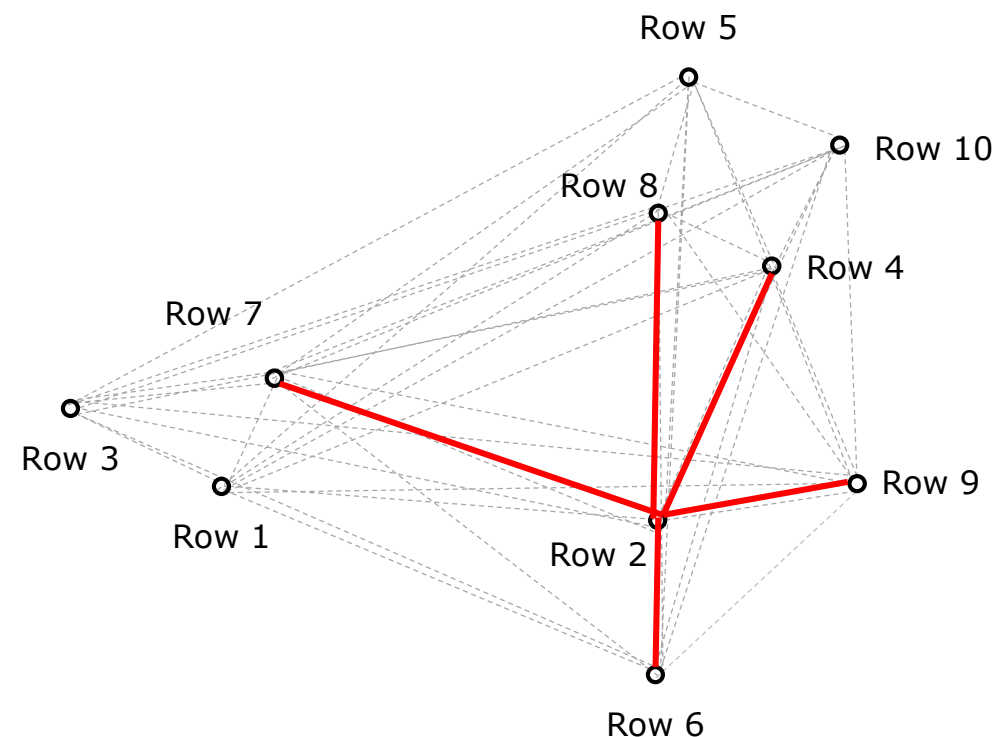
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Row 3	0.8	0.2		0.1	0.1	0.1	0.8	0.1	0.1	0.1
Row 4	0.2	0.4	0.1		0.7	0.5	0.2	0.9	0.5	0.9
Row 5	0.1	0.1	0.1	0.7		0.1	0.1	0.0	0.3	0.9
Row 6	0.2	0.9	0.1	0.5	0.1		0.2	0.2	0.3	0.1
Row 7	0.9	0.4	0.8	0.2	0.1	0.2		0.1	0.1	0.1
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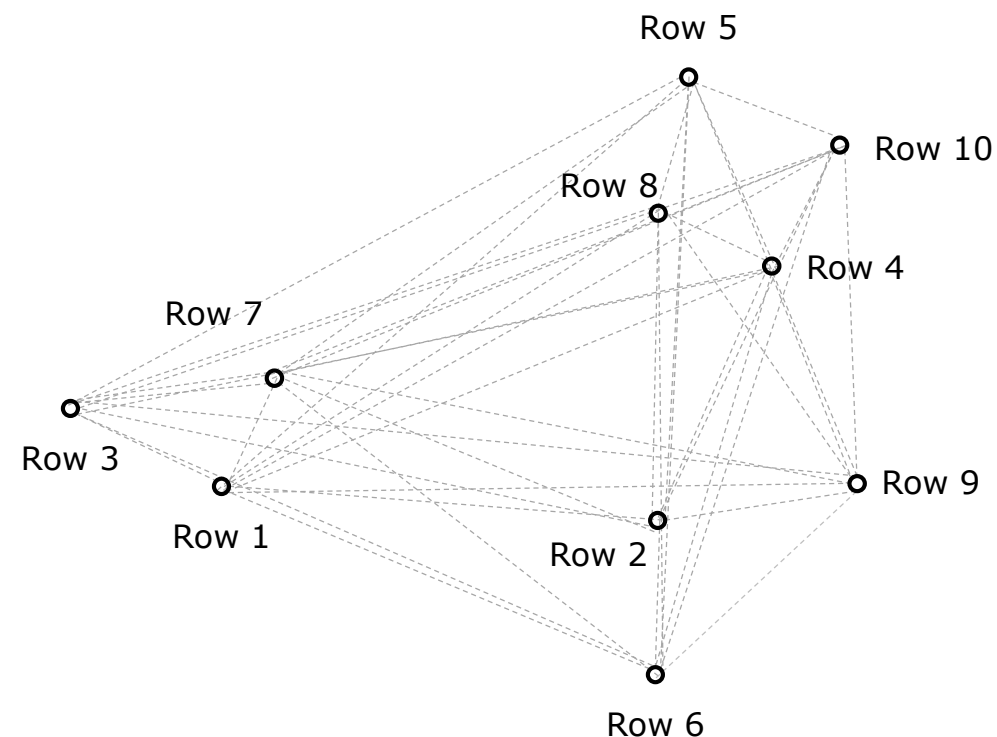
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Row 4	0.2	0.4	0.1		0.7	0.5	0.2	0.9	0.5	0.9
Row 5	0.1	0.1	0.1	0.7		0.1	0.1	0.0	0.3	0.9
Row 6	0.2	0.9	0.1	0.5	0.1		0.2	0.2	0.3	0.1
Row 7	0.9	0.4	0.8	0.2	0.1	0.2		0.1	0.1	0.1
Row 8	0.2	0.4	0.1	0.9	0.0	0.2	0.1		0.3	0.8
Row 9	0.1	0.6	0.1	0.5	0.3	0.3	0.1	0.3		0.4
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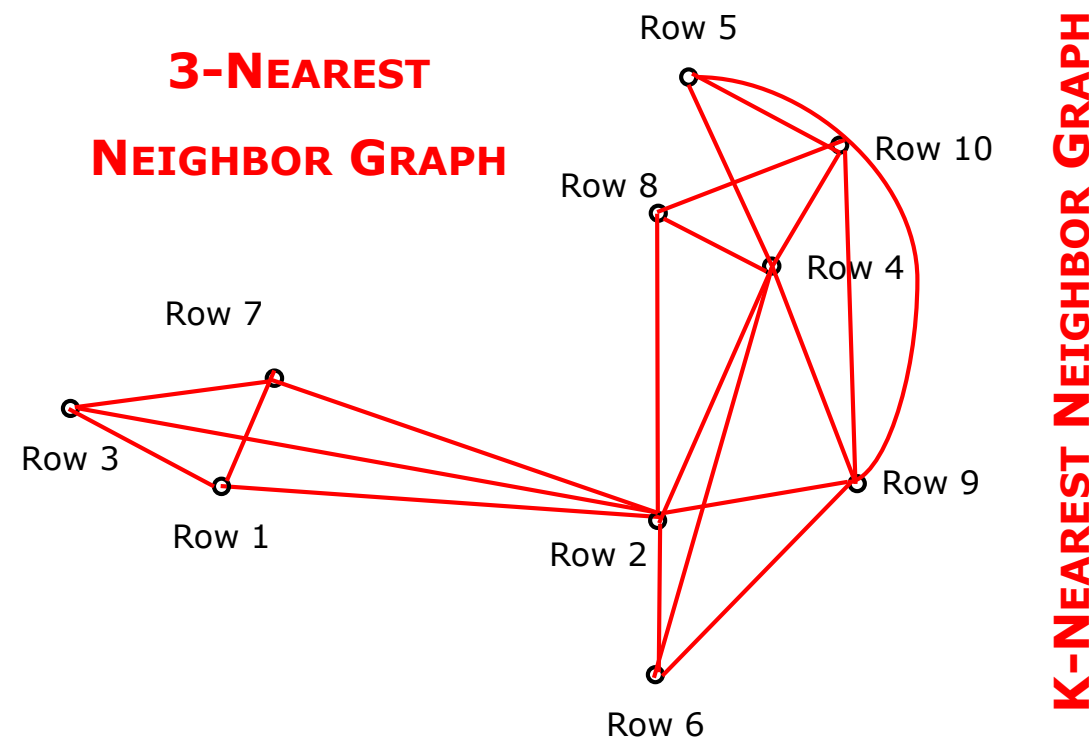
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Row 3	0.8	0.2		0.1	0.1	0.1	0.8	0.1	0.1	0.1
Row 4	0.2	0.4	0.1		0.7	0.5	0.2	0.9	0.5	0.9
Row 5	0.1	0.1	0.1	0.7		0.1	0.1	0.0	0.3	0.9
Row 6	0.2	0.9	0.1	0.5	0.1		0.2	0.2	0.3	0.1
Row 7	0.9	0.4	0.8	0.2	0.1	0.2		0.1	0.1	0.1
Row 8	0.2	0.4	0.1	0.9	0.0	0.2	0.1		0.3	0.8
Row 9	0.1	0.6	0.1	0.5	0.3	0.3	0.1	0.3		0.4
Row 10	0.1	0.2	0.1	0.9	0.9	0.1	0.1	0.8	0.4	

K = 3



SPARSIFIED PROXIMITY GRAPH

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ADVANTAGES OF SPARSIFICATION

- **REDUCTION OF THE DATA SIZE;** eliminates most of the entries of the proximity matrix
- **CLUSTERING MAY WORK BETTER;** the impact of noise and outliers is reduced due to the nearest neighbor principle, i.e., the nearest neighbors of an object tend to belong to the same class (cluster) as the object itself
- **GRAPH PARTITIONING ALGORITHMS CAN BE USED;** a considerable amount of work on heuristic algorithms for finding min-cut partitionings of sparse graphs

